

Antibiotics

These wonder drugs are one of medicine's most important weapons in the fight against disease-causing bacteria. But the first antibiotic was discovered entirely by accident.

Stopping DISEASES in their tracks



Fleming's fluke

By the 19th century, scientists had figured out that many diseases are caused by **microscopic living things** called bacteria. In 1928, Scottish bacteriologist **ALEXANDER FLEMING** noticed that one of his petri dishes containing *Staphylococcus* bacteria had become contaminated with a blue mold, which seemed to have wiped out the bacteria. Fleming realized that the mold contained a bacteria-killing chemical, which he called **PENICILLIN**.

Florey and Chain

Fleming had chanced upon the first antibiotic, but **isolating penicillin** in quantities large enough to treat people took more than a decade. A team in Oxford, England, led by Australian scientist **Howard Florey** used hospital bedpans and cow-milking equipment to produce enough penicillin to run **MEDICAL TRIALS** on humans in 1941. They proved successful in fighting a range of infections and diseases.

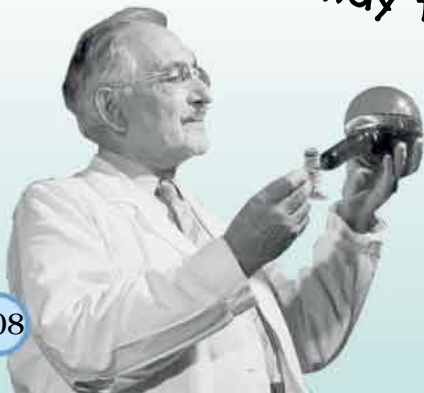
Ernst Chain proved penicillin's success in fighting bacteria in mammals through experiments with mice.



By the way...

I was a key member of Howard Florey's team. Along with Florey and Alexander Fleming, I was awarded the 1945 Nobel Prize for Medicine.

It paved the way for...



STREPTOMYCIN was first made in 1943 by American scientist **SELMAN WAKSMAN** from bacteria found in soil. It was the first antibiotic remedy for tuberculosis.

TETRACYCLINE was first discovered in 1945 and then made in labs by American drug company Pfizer in the early 1950s. It became the **MOST PRESCRIBED** antibiotic in the United States.

